

Operation	Hardware type	Hardware length	Hops
Transactions ID			
Seconds		Flags	
Client IP address			
Your IP address			
Server IP address			
Router / Gateway IP address			
Client hardware address (16 bytes)			
Server host name (64 bytes)			
File (128 bytes)			
Operation			

Figure 3: A DHCP message sent using UDP

What cellular technology, Global System for Mobile (GSM) communication or cellphone do for telephony, mobile IP will do for TCP/IP-based mode of data transport. IP Internet has become ubiquitous in wireline environments. Support of mobility is already available today in some forms using satellite access to an Internet service provider (ISP). But satellites provide access to just one ISP at a time, and only as long as the user is within the satellite's footprint.

Mobile IP provides a platform for carriers to offer assured and continuous access to the Internet or a private IP-based network. IP datagram is a packet with data plus header fields of different layers of the Internet architecture. This is like a letter that contains a message plus other data like sender and receiver names, addresses, pin codes, etc.

An IP datagram carries both source and destination addresses of data. Thus, if source or destination changes, the connection becomes void until and

unless the changed IP address is notified to all users.

In conventional wired Internet connectivity, when a user gets connected to the Internet, the point of attachment is made through an address known as IP address. This IP address is attached to a fixed plugged-in position.

In wireless connectivity, when a user is connected, he or she is connected to an IP address that is valid over the coverage area cell of wireless connectivity. The mobile user changes location dynamically with time. Thus, plugged-in fixed IP address of the wired Internet will be of no use to support connectivity to the mobile user.

On the other hand, there is no guarantee that mobility will be limited to the coverage area in case of wireless connectivity. IP address has to change when the mobile user moves away from own coverage or cell area to remain in connectivity.

One simple solution is to not assign a fixed IP address. There is a requirement to dynamically change the IP address. A new protocol called DHCP resolves the stated issue. DHCP allows the intended user to obtain a temporary IP address dynamically. Like when one goes on deputation, one is allotted a temporary office/residential address.

This is a simple management solution. But a temporary address has two major disadvantages: temporary address notification and system upgradation. Other nodes cannot easily originate transmissions to the mobile node (MN) as these do not know about its new IP address. And each time a mobile computer obtains a new IP address, Internet client software applications must be restarted.

Again, temporary addresses do not let a mobile user roam seamlessly from one IP subnet area to another. This creates a bottleneck, especially for wireless local area networks (LANs). As an example, a doctor with a wireless LAN connection needs to maintain IP connections throughout the hospital as he or she accesses patient records. This is where the need for a different solution by mobile IP arises.

Mobile IP enables mobile users to stay connected to the Internet regardless of their location and without changing their IP address. It allows them a fixed IP address for being connected to any IP subnet and immediately be reachable from the Internet.

Mobile IP is an extension of conventional IP and makes mobility transparent to higher layers of the Internet architecture. In fact, it is not just wireless connectivity, it is mobile connectivity that supports seamless machine mobility with the support of Internet+DHCP.

Relational feature of DHCP, ICMP, ARP over basic TCP/UDP is portrayed in Figure 4. Internet Engineering Task Force (IETF) defines mobile computer as a device used to carry data as users move from one network to another while maintaining their permanent IP address.

Request for Comments (RFC) defines it as, "Mobile IP is an enhancement of the Internet Protocol that adds mechanisms for forwarding Internet traffic to mobile devices when they are connecting through other than their home network." Mobile IP

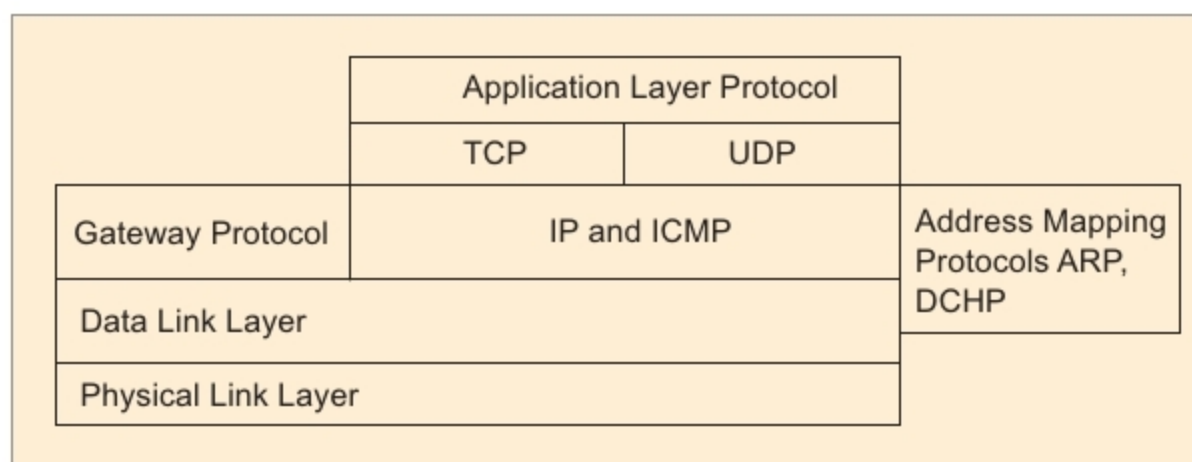


Figure 4: TCP/IP (Internet) model [TCP = Transmission Control Protocol, UDP = Users' Datagram Protocol, ICMP = Internet Control Message Protocol, ARP = Address Resolution Protocol, DHCP = Dynamic Control Host Protocol]